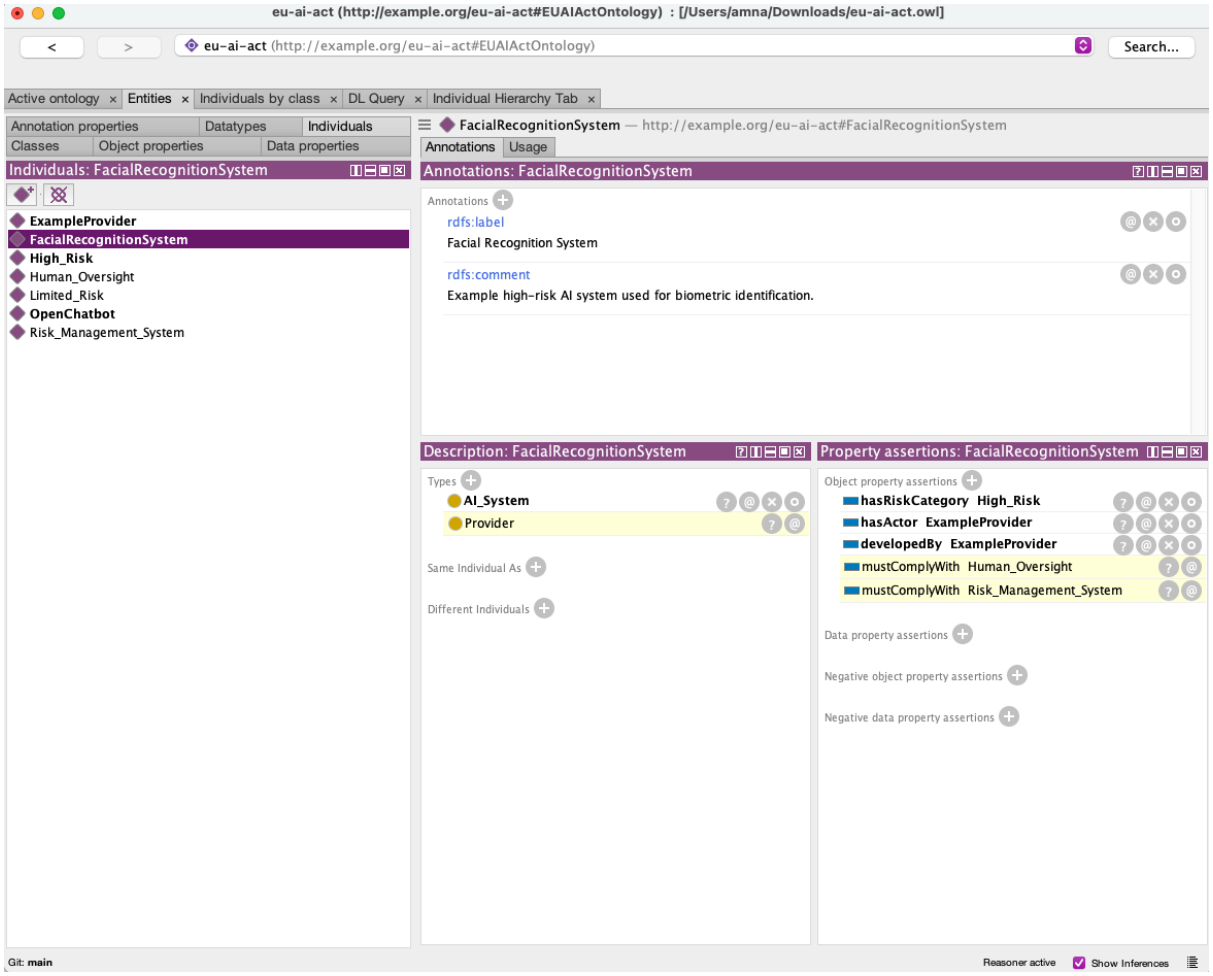
Basic validation

The ontology was validated using the **HermiT** to ensure logical consistency.
Preliminary evaluation approaches include:

- checking logical consistency of the ontology
- verifying coverage of core regulatory concepts
- assessing competency question support through SPARQL queries.



SPARQL Query & Update

EU_AI_ACT | en

Editor only | Editor and results | Results only

Unnamed

```
PREFIX : <http://example.org/eu-ai-act#>
SELECT ?system
WHERE {
  ?system a :AI_System ;
          :hasRiskCategory :High_Risk .
}
```

Table | Raw response | Pivot Table | Google Chart | Download as

Filter query results | Compact view | Hide row numbers

Showing results from 0 to 1 of 1. Query took 0.1s, moments ago.

	system
1	:FacialRecognitionSystem

SPARQL Query & Update

EU_AI_ACT | en

Editor only | Editor and results | Results only

Unnamed

```
PREFIX : <http://example.org/eu-ai-act#>

SELECT ?provider ?obligation
WHERE {
  ?provider a :Provider ;
            :hasObligation ?obligation .
}
```

Table | Raw response | Pivot Table | Google Chart | Download as

Filter query results | Compact view | Hide row numbers

Showing results from 0 to 1 of 1. Query took 0.1s, moments ago.

	provider	obligation
1	:ExampleProvider	:Risk_Management_System

SPARQL Query & Update

EU_AI_ACT | en

Editor only | Editor and results | Results only

Unnamed

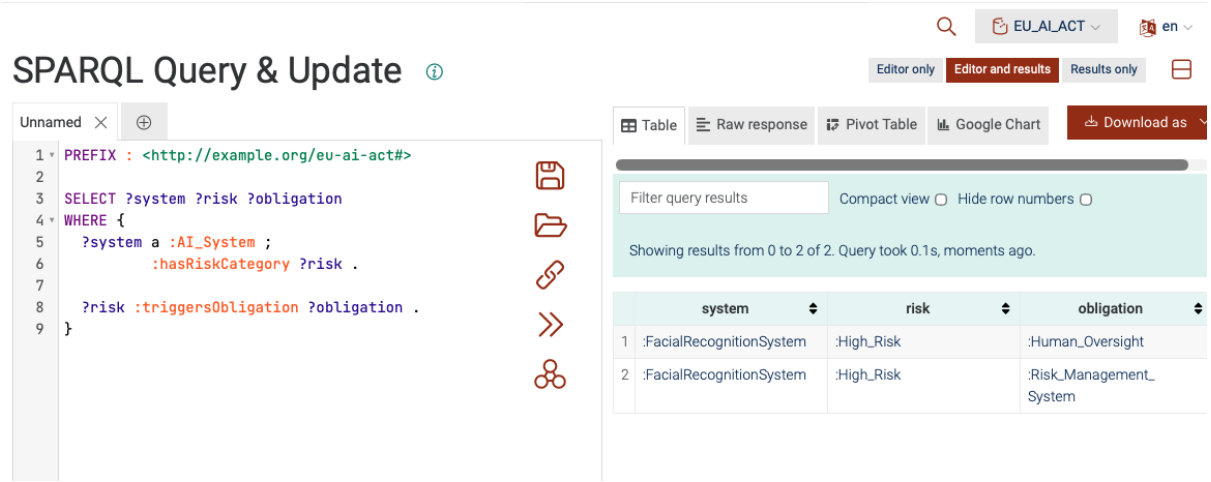
```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX : <http://example.org/eu-ai-act#>
SELECT ?practice
WHERE {  ?practice rdfs:subClassOf
:Prohibited_AI_Practice .}
```

Table | Raw response | Pivot Table | Google Chart | Download as

Filter query results | Compact view | Hide row numbers

Showing results from 0 to 3 of 3. Query took 0.1s, moments ago.

	practice
1	:RealTime_Biometric_Identification
2	:Social_Scoring
3	:Subliminal_Manipulation



Preliminary Evaluation Metrics and Methods

Several preliminary metrics and evaluation methods can be used to assess the quality and usefulness of the ontology.

Logical consistency
The ontology can be evaluated using OWL reasoners such as **HermiT** to detect logical inconsistencies, unsatisfiable classes, or conflicting class hierarchies.

Competency question coverage
The ontology can be tested against the defined competency questions by executing SPARQL queries. Successful retrieval of expected results indicates that the ontology adequately captures the intended domain concepts and relationships.

Structural quality metrics
Basic ontology metrics such as number of classes, object properties, individuals, and class hierarchy depth can be used to evaluate structural completeness and modelling complexity.

Conceptual coverage
The ontology can be assessed by verifying whether it represents the main regulatory concepts of the EU AI Act, including actors, AI systems, risk categories, obligations, and prohibited practices.

These preliminary evaluations help ensure that the ontology is logically consistent, conceptually coherent, and capable of supporting basic regulatory queries.